SESSION 1.2 – ACTIVITY

Foundations of AI & History of AI

Activity: Examine the AI literature to discover whether the following tasks can currently be solved by computers.

Task	Can computers do it?	Explanation	Conclusion
1. Playing a decent game of table tennis	Yes	Computers and AI-controlled robots can play table tennis at a good level. Using cameras, sensors, motion tracking, and machine learning, a robot can detect the ball, predict its path, and move the paddle quickly. However, human-level performance in all real-world situations is still challenging.	Currently achievable with AI/computers.
2. Discovering and proving a new mathematical theorem	Yes, to some extent	AI systems and automated theorem provers can discover useful mathematical patterns and prove many theorems. Modern AI can also assist mathematicians in finding new conjectures and proofs. For very difficult or highly creative mathematics, human guidance is still often required.	Achievable, but not fully independent in every case.
3. Writing an intentionally funny story	Yes	Generative AI can write humorous stories, jokes, dialogues, and creative fiction. It can imitate different writing styles and use wordplay. Since humor depends on culture and personal taste, the result may not always be funny to every reader.	Currently achievable with AI/computers.

4. Translating spoken English to spoken French	Yes	Current speech-recognition and machinetranslation systems can convert spoken English into French and generate the translated speech almost instantly. This is already available in translation applications and voice-assistant systems. Accuracy may decrease with strong accents, background noise, or technical vocabulary.	Currently achievable with AI/computers.
5. Performing a complex surgical operation	Partly	Robotic surgical systems can assist surgeons in complex operations with high precision, and AI can support planning, imaging, and decision-making. Some limited procedures can be automated, but most complex surgeries still require trained surgeons to supervise and control the operation for safety.	AI can assist, but complete autonomous performance is still limited.

Overall Conclusion: Most of these tasks can already be performed by computers or AI systems. Creative writing and translation are widely available, mathematical discovery is increasingly supported by AI, table-tennis robots can perform well in controlled settings, and surgical AI is mainly used as an assistive technology rather than as a fully independent surgeon.

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